Matching

CS1200: Introduction to Algorithms and their Limitations
Lecture 14

Fall 2026

Harvard SEAS

Recap: when does greedy work?

Def. 14.1. An independent set in G is a set of vertices with no edge between any two of them.

problem	greedy	best known
Interval Scheduling (§14.3)	optimal, sorted by end time	$O(n \log n)$
Independent Set (§14.1)	can be far from optimal	roughly 1.2^n
Graph 2-Coloring (§13.2)	optimal, in BFS order	$O(n + m)$
Graph 3-Coloring (§14.2)	no guarantee	roughly 1.33^n
Maximum Matching (today)	fails...	...but can be repaired

Kidney exchange

- US kidney waiting list: about 90,000 patients; a little over 27,000 transplants a year; average wait about 3.6 years.
- A donor gives one kidney, and only to a *compatible* patient (blood type, HLA type).

Model. A compatibility graph: a vertex for each donor and each patient, an edge for each compatible donor–patient pair.

- Before: edges meant *conflict* (registers, intervals). Now an edge means _____.
- Constraints: each donor gives to at most one patient; each patient receives at most one kidney.
- What kind of graph is it? _____

Goal: as many transplants as possible.

Definition 15.1: matching

Def. 15.1. A matching in $G = (V, E)$ is a set $M \subseteq E$ such that every vertex is incident to at most one edge of M.
Equivalently: _____

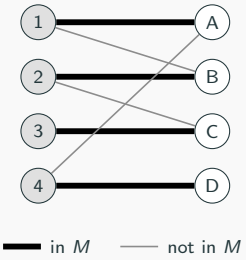
v is matched by M if some edge of M contains it; otherwise unmatched.

Maximum Matching

$\mathcal{I}$: graphs $G = (V, E)$
$\mathcal{O}$: sets of edges M
$f(G) = \{M : M \text{ a matching in } G \text{ of maximum size}\}$

At right: $|M| = 4$. Could any matching be bigger?

These definitions are for *any* graph G, not just bipartite.



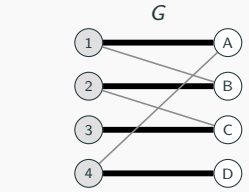
Matching is independent set in disguise (§15.4)

Line graph $L(G)$: one vertex per *edge* of G; two are adjacent if the edges share an endpoint.

Matchings in $G =$ _____

So Maximum Matching reduces to Independent Set.
Is that a good way to solve it?

Like interval scheduling: an *easy special case* of a hard problem.



Greedy fails

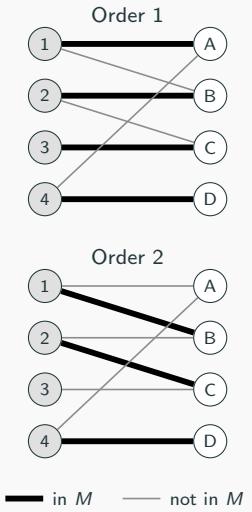
Greedy: start with $M = \emptyset$; go through the edges in some order; add an edge if it shares no endpoint with M.

Order 1:
$\{1, A\}, \{1, B\}, \{2, B\}, \{2, C\}, \{3, C\}, \{4, D\}, \{4, A\}$.
$|M| = 4$: maximum.

Order 2: $\{4, D\}, \{2, C\}, \{1, B\}$, then the rest.
Every remaining edge touches a matched vertex: maximal (can't add an edge), not maximum.

What went wrong?

Repair: allow swaps. But which swaps, so that we always make progress?



Definition 15.2: alternating walks and augmenting paths

Let M be a matching in G.

An alternating walk (w.r.t. M) is a walk $v_0, v_1, \ldots, v_\ell$ whose edges alternate between _____

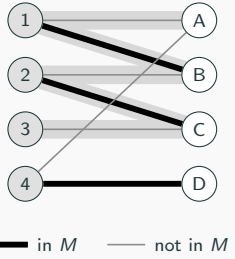
An augmenting path is an alternating walk with $\ell \geq 1$ such that

- v_0 and v_ℓ are _____
- all of its vertices are distinct.

At right: $3, C, 2, B, 1, A$.
Alternating? _____
Augmenting? _____

Any other augmenting paths?

_____



Flipping an augmenting path (Lemma 15.7)

Flip the path: its edges in M leave M; its edges not in M join M.

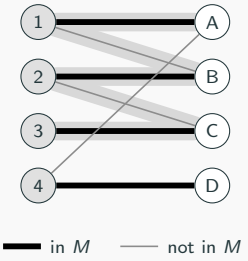
Lemma 15.7. If P is an augmenting path w.r.t. M, flipping P gives a matching M' with $|M'| = |M| + 1$, in time $O(n)$.

Why $+1$? _____

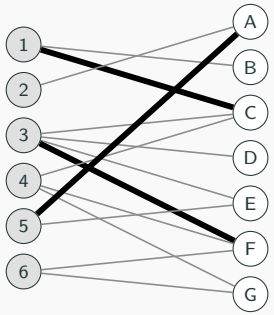
Why still a matching?

_____

Here: size 3 $\rightarrow$ size 4, the maximum.



In groups: alternating? augmenting?



$M = \{\{1, C\}, \{3, F\}, \{5, A\}\}$

For each sequence: is it an alternating walk? An augmenting path?

sequence	alternating walk?	augmenting path?
$2, A, 5$		
$G, 4$		
$3, F, 4, C, 3$		
$1, C, 6$		
$6, F, 3, C, 1, B$		

Berge's Theorem (Theorem 15.3)

Theorem 15.3 (Berge). A matching M in a graph G is not of maximum size if and only if G has an augmenting path w.r.t. M.

$\Leftarrow$: _____

$\Rightarrow$, **proof idea.** Take a bigger matching M^*. Keep only the edges in exactly one of M, M^*: $M \triangle M^*$.

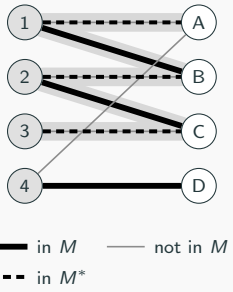
- Every vertex has degree ≤ 2 in it:

  _____

- So it splits into paths and cycles, alternating M / M^*.
- Cycles have as many M edges as M^* edges.
  $|M^*| > |M|$, so some path has more M^* edges:

  _____

Holds for *every* graph, not just bipartite ones.



$\{4, D\}$ is in both, so it drops out.

The algorithm (Algorithm 15.1)

MaxMatchingAugPaths(G)

Input: a bipartite graph G

0 remove isolated vertices from G
1 let V_0, V_1 be the two sides (a 2-coloring)
2 $M = \emptyset$
3 **repeat**
4 $\quad U_0, U_1 =$ vertices of V_0, V_1 unmatched by M
5 $\quad$ find an augmenting path P from U_0 to U_1 (or $\perp$)
6 $\quad$ **if** $P \neq \perp$: flip P (Lemma 15.7)
7 **until** $P = \perp$
8 **return** M

Correct: it stops only when there is no augmenting path, so by

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Left to do: line 5. How do we *find* an augmenting path?

Finding an augmenting path: shortest paths (Lemma 15.6)

Build a *directed* graph G':

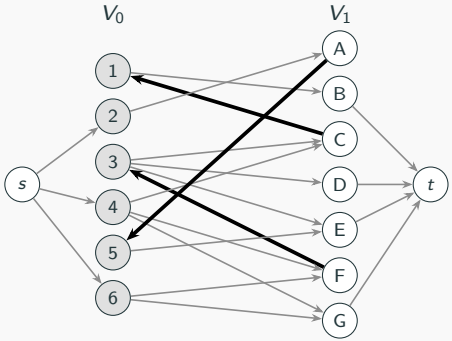
1. edges not in M: point $V_0 \to V_1$;
2. edges in M: point $V_1 \to V_0$;
3. new s, with $s \to u$ for each $u \in U_0$;
4. new t, with $u \to t$ for each $u \in U_1$.

Why does an $s \to t$ path alternate?

BFS from s:

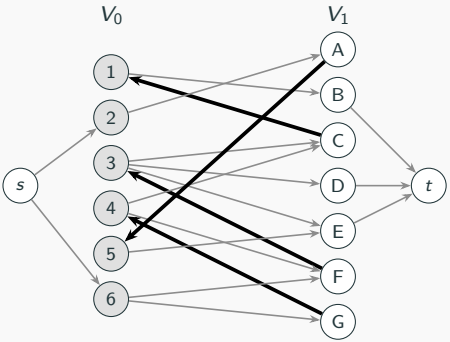
Drop s, t: augmenting path

What does the algorithm do on the first round if the matching is empty?



$M = \{\{1, C\}, \{3, F\}, \{5, A\}\}$; $U_0 = \{2, 4, 6\}$, $U_1 = \{B, D, E, G\}$

In groups: the next augmenting path



$M = \{\{1, C\}, \{3, F\}, \{4, G\}, \{5, A\}\}$

1. What changed in G' since the last slide?

   _____

2. BFS from s. Frontiers:
   $F_1 =$ _____, $F_2 =$ _____,
   $F_3 =$ _____, $F_4 =$ _____
   Distance to t: _____

3. A shortest path, and the new M:

   _____

   _____

Runtime (Theorem 15.8)

Theorem 15.8. Maximum Matching on bipartite graphs can be solved in time $O(mn)$.

one round:	build G' and run BFS	_____
	flip the path	_____
number of rounds		_____
total		_____

No isolated vertices (line 0) $\Rightarrow n \leq 2m \Rightarrow O(n(n+m)) = O(mn)$.

Not bipartite? Berge still holds, but we're relying on bipartiteness to find an augmenting path. Efficient algorithms exist (Edmonds' "blossoms"), with similar runtime and much more work.